Sephora Product Reviews

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Abstract

This project explores customer reviews and product data from Sephora to uncover what drives positive skincare experiences and to build models that can predict ratings and sentiment. We start by cleaning and exploring the data-verifying rating ranges, removing duplicates, and normalizing review text-then analyze key patterns across brands, prices, and user traits such as skin type. Next, we create features from both text and metadata, combining lexical representations (TF-IDF) and transformer embeddings (DistilBERT) with product-level factors (like price and brand) and reviewer context (e.g., review length or skin tone). For modeling, we compare strong linear baselines such as Logistic Regression and SVM with a fine-tuned transformer for sentiment classification. Model performance is assessed using stratified cross-validation with accuracy, F1, and AUC metrics, followed by detailed error and bias analysis to understand where the models struggle. The results will offer insights at both brand and attribute levels and deliver a reproducible pipeline—complete with code, figures, and documentation—in a public GitHub repository.

Literature Review

Recent advances in sentiment analysis and recommendation systems have opened new opportunities for their application in the cosmetics domain, while also revealing important research gaps. **Gajula et al. (2025)** present a comprehensive survey of sentiment-aware recommender systems in e-commerce, emphasizing that integrating textual sentiment with numerical ratings enhances both accuracy and explainability. **Xu (2024)** introduces a hybrid approach that combines dependency parsing with sentiment polarity to extract more robust attribute labels from product reviews, including those in cosmetics. **Huang et al. (2020)** propose a weakly-supervised joint embedding framework for aspects and sentiments, effectively reducing dependence on large annotated datasets. Similarly, **Agarap (2018)** demonstrates that bidirectional LSTM models outperform traditional classifiers in identifying sentiment in clothing reviews, highlighting that numerical ratings often serve as imprecise proxies for actual sentiment.

Within the cosmetics sector, several domain-specific studies have applied Support Vector Machines (SVM) and Naïve Bayes to analyze beauty product reviews—such as those from the Sephora dataset—achieving accuracies between approximately 80% and 90%. More recent transformer-based hybrid architectures further improve classification and aspect extraction performance when tailored to beauty and skincare products. Collectively, these studies show that while traditional machine learning models remain competitive, deep learning and transformer-based models are advancing the state of the art.

A key insight across the literature is that Aspect-Based Sentiment Analysis **(ABSA)** plays a pivotal role in the cosmetics domain, where user reviews frequently address multiple product attributes such as scent, texture, and efficacy. However, few studies integrate consumer attributes (e.g., skin type, skin tone) and product metadata (e.g., brand, price) into the sentiment modeling framework. Addressing this research gap, the present study proposes a cosmetics-specific ABSA and rating prediction model that jointly leverages review text, consumer characteristics, and product-level metadata for more comprehensive and personalized sentiment insights.

Data Description

Expletory Data Analysis (EDA) Values: Description

The dataset includes roughly 50,000 Sephora product reviews, covering about 1,100 products from over 120 brands. Each entry contains a 1–5 star rating, review text and title, brand and product details, price, and user attributes such as skin type, skin tone, and eye or hair color, along with user feedback metrics.

During data cleaning, we removed invalid ratings, eliminated duplicates, and standardized the review text for consistency. After preprocessing, the final dataset consisted of approximately 49,972 reviews. Overall, ratings were positively skewed, with most users giving high scores. The median product price was around $45, while luxury items reached up to $1,900. A few notable outliers included 1,111 reviews for products priced above $300, seven unusually long reviews (over 500 tokens), and 353 reviews with feedback counts exceeding 100.

Correlations: Ratings correlate positively with helpfulness (+0.175) and review length (~+0.03 to +0.04), and negatively with feedback counts (−0.10 to −0.17). Limitations: ingredient lists are not present, limiting direct ingredient-sentiment analysis. Proxy features (e.g., keywords in product names like 'retinol' or 'SPF') will be extracted.

As shown in Figure 1, most products received very high ratings, with a strong concentration at five stars. Lower ratings (between one and three stars) appeared far less frequently, producing a clear right-skewed distribution. This pattern suggests that Sephora customers generally express high satisfaction-though it may also reflect a degree of rating bias common in e-commerce platforms.

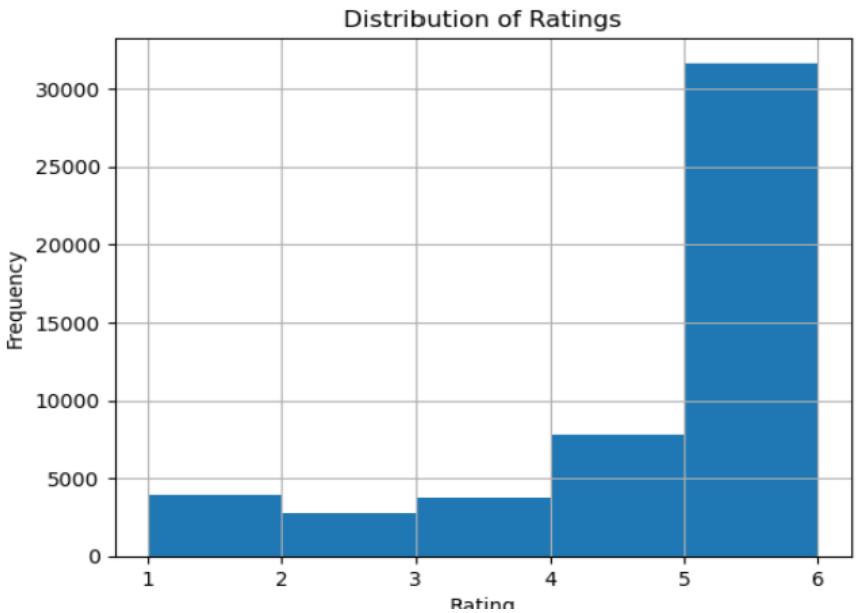


Figure 1: Frequency vs Rating.

The scatter plot (**Figure2**) shows the relationship between **review length** and **rating**. Most reviews are short, under 100 words, and clustered around **4–5 stars**, suggesting that users tend to give high ratings regardless of review length. There is no clear correlation, indicating that longer reviews do not necessarily correspond to higher or lower ratings.

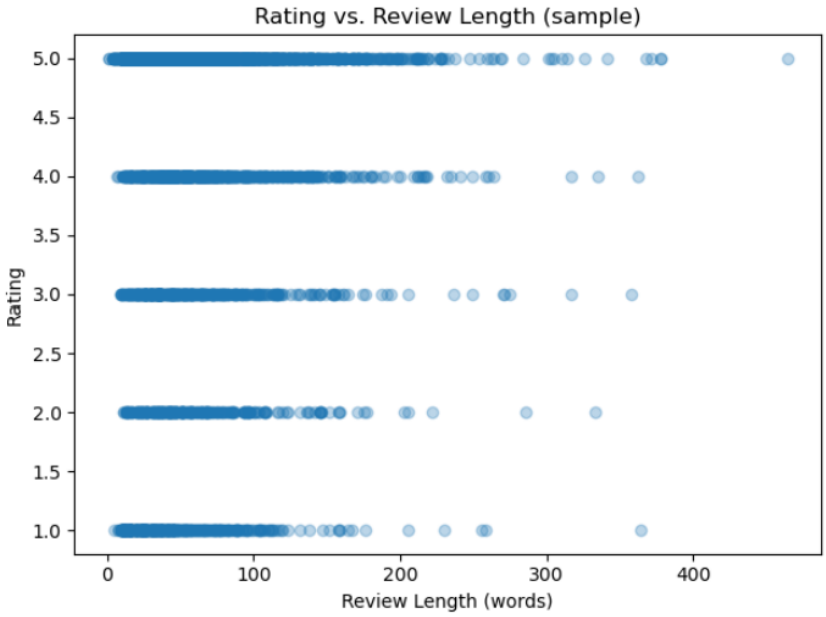


Figure 2: Rating vs Review Length.

Figure 3 shows the distribution of numeric features in the Sephora dataset. Most variables, such as **rating**, **helpfulness**, and **price_usd**, have compact ranges with several outliers, indicating skewed distributions. The **Unnamed: 0** column displays a much wider spread, confirming it's likely just an index field rather than meaningful data. Overall, the plot highlights variation across features and the presence of outliers that may require normalization or scaling.

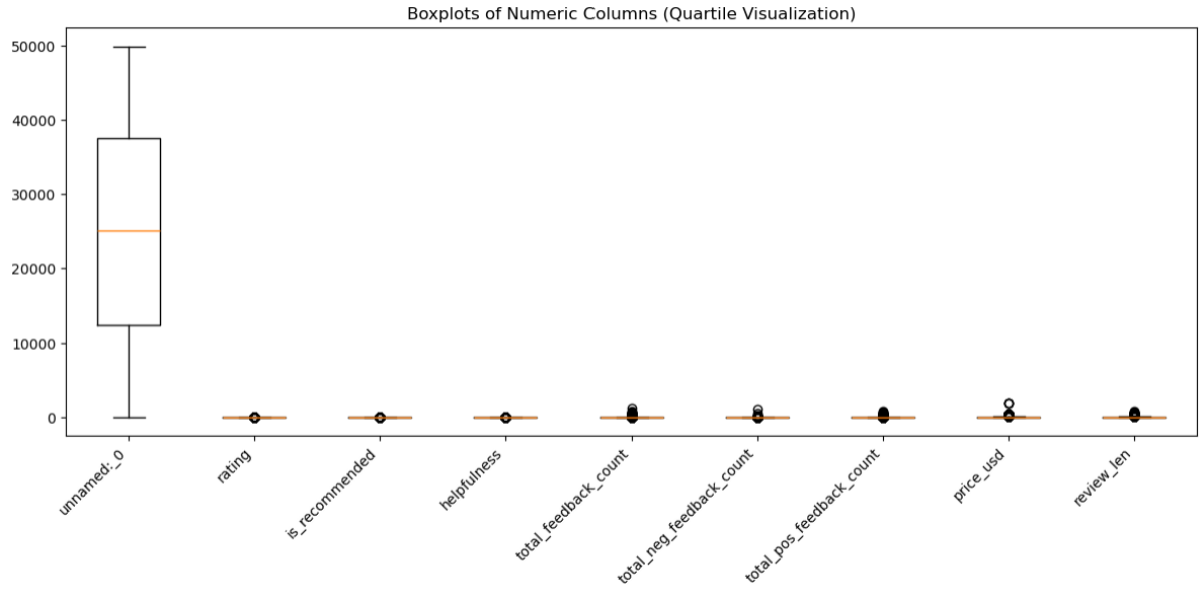


Figure 3: Boxplots of the columns

Figure 3 shows the distribution of scaled product ratings in the Sephora dataset. Most ratings are concentrated near **1.0**, indicating that users predominantly give high scores. This strong right skew suggests a **positive bias** in customer feedback, where negative or moderate ratings are relatively rare.

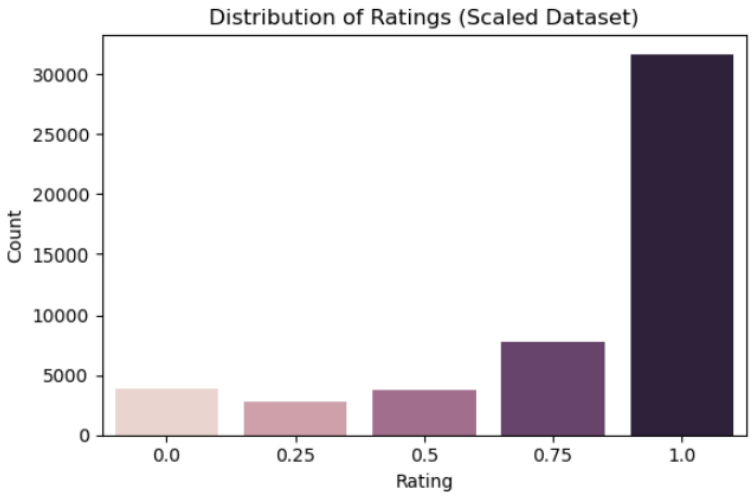


Figure 4: Boxplots of the columns

Figure 5 shows strong positive correlation (**0.87**) between **rating** and **is_recommended**, meaning higher-rated products are more likely recommended.

Feedback-related variables (**total_feedback_count**, **pos**, **neg**) are also highly correlated, indicating consistent user engagement patterns.

In contrast, **price** and **helpfulness** show weak or no correlation with ratings, suggesting they have little influence on user scores.

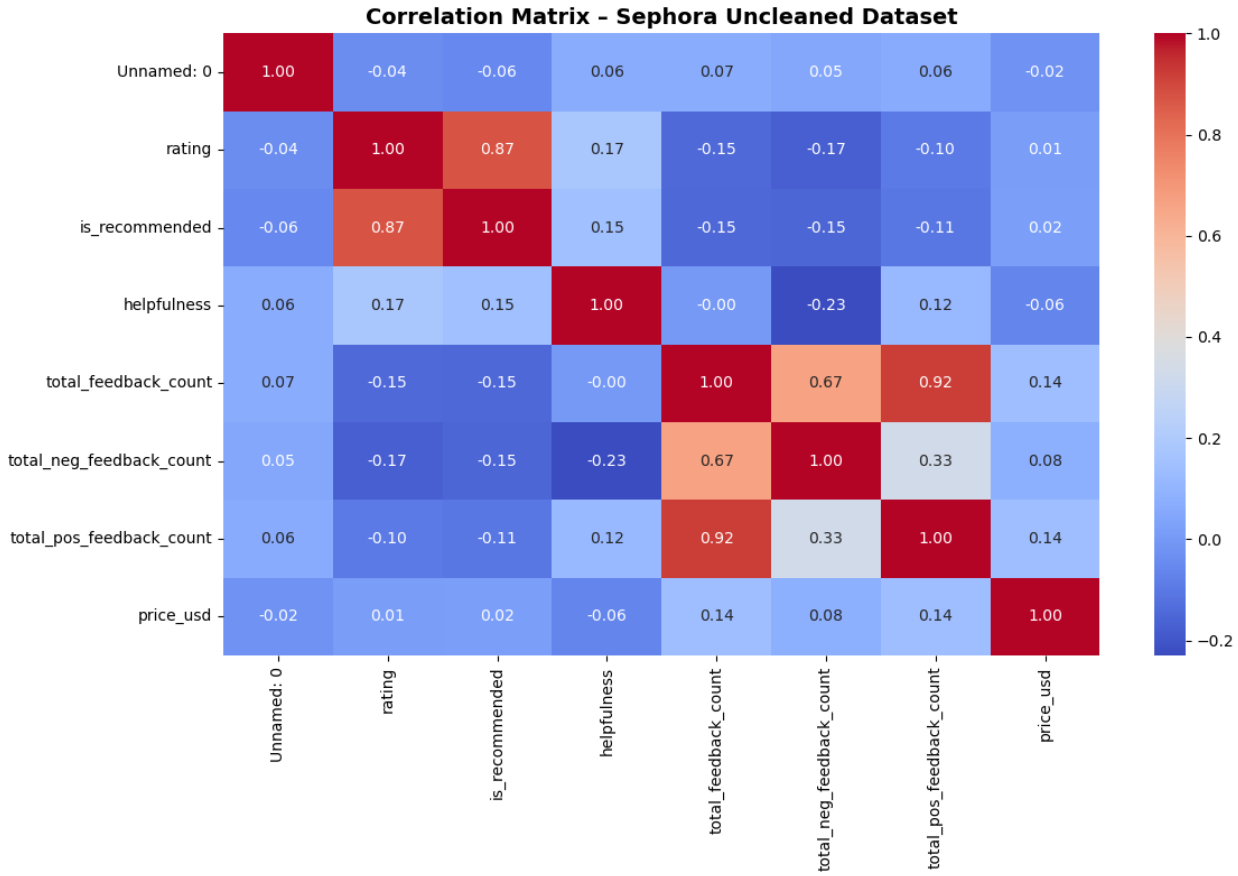


Figure 5: Correlation Matrix

Approach

Project Methodology Flowchart

The diagram (**Figure 6**) illustrates the data science pipeline for the cosmetics review analysis project, showing the sequential flow from data collection to final deliverables.

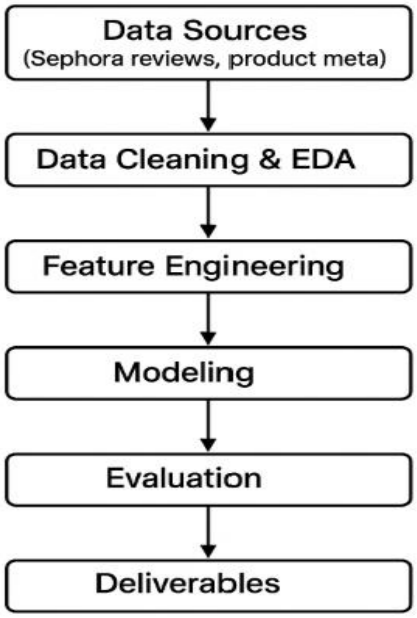


Figure 6: End-to-end workflow from Data Sources through Deliverables.

Research Questions

1. How do sentiment and aspect-level opinions vary across cosmetics categories?
2. Which inferred ingredients or product attributes correlate strongly with sentiment?

3. Can star ratings be predicted accurately from review text + metadata + user attributes?

Methodological Steps:

1. Text + Feature Engineering: Preprocessing (lowercasing, tokenization), engineered features (review length, punctuation density, all-caps ratio), and lexicon/syntax-based extraction for attributes.
2. EDA: rating distributions, brand-level comparisons, attribute-level sentiment heatmaps, clustering by attributes.
3. Modeling: Baselines (TF-IDF + LogReg, SVM, Naïve Bayes); Advanced (BERT/DistilBERT fine-tuning). Hybrid models incorporate metadata.
4. Evaluation: 5-fold cross-validation, macro-F1, per-class metrics, temporal validation, ablation studies, and error analysis.
5. Contribution: Extends research by integrating consumer attributes, metadata, and inferred ingredients into cosmetics-focused ABSA and rating prediction.
6. Reproducibility: notebooks, configs, dataset versions, and model artifacts will be shared.

GitHub Repository Link

All code, preprocessing scripts, and results will be maintained in a GitHub repository. Please visit the repository at [https://github.com/akazemza/Sephora-BigData-Project]. This ensures full reproducibility and transparency of the workflow, aligned with best practices.

References

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